

FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10 - 1.1 PRESSURE: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

You have finished all 9 lessons on pressure. Now you will write your own Final Explanation of the driving question: Why does a sharp knife cut better than a blunt knife when using the same force? Answer each section in the space below it. Use numbers with their units, and use evidence from your lessons and your Driving Question Board. Use $g = 10 \text{ N/kg}$. Remember to say what is the same and what is different between the two knives. Write in full sentences.

Part 1: Claim and Evidence from the Kitchen

Mama Wanjiru in Kisumu pushes with the same force on a sharp knife (edge 0.1 mm thick) and a blunt knife (edge 1 mm thick). Both knives are the same size and shape and touch the food along a length of 5 cm. (a) Write a claim that answers the driving question. (b) List what is the same and what is different in this situation. (c) Give the force value and what happened with each knife.

Write your answer here:

Part 2: Calculating the Pressure Under Each Knife

Use $P = F/A$ to explain the knives with numbers. (a) Change each edge thickness to metres and find the contact area of each knife (contact length 0.05 m). (b) Calculate the pressure under each knife and give your answers in MPa. (c) Compare the two areas and the two pressures. How many times larger or smaller is each? (d) Tomato skin and nyama choma give way at about 2 MPa. Use this value to explain what happens with each knife.

Write your answer here:

Part 3: Testing the Idea with Other Evidence

Use evidence from your earlier lessons to show that the area-pressure idea works beyond the knives. (a) In the brick-on-flour test, the same 2 kg brick (weight 20 N) was placed on its large face (0.02 m²) and its small face (0.005 m²). Calculate the pressure for each face and say which face sank deepest. (b) A camel (4,000 N, total foot area 0.2 m²) and a 60 kg student (600 N, total shoe area 0.04 m²) stand on sand. Calculate each pressure and explain why the much heavier camel does not sink much more than the student.

Write your answer here:

Part 4: Pressure in Fluids and Its Uses

The knife idea can be extended to fluids. (a) Water in a dam is 30 m deep. Use $P = \rho gh$ (water density 1,000 kg/m³) to calculate the pressure at the base and at 10 m depth. Explain why a dam wall is built thicker at the bottom. (b) In the glass-with-cardboard test, the glass mouth has area 0.004 m², the atmospheric pressure is 101 kPa, and the water weighs 2.5 N. Calculate the upward force of the air and explain what it shows. (c) In a hydraulic jack, a 50 N effort acts on a piston of 5 cm² and the large piston has 100 cm². Find the pressure and the load force, and say how the idea differs from the knives.

Write your answer here:

Part 5: Final Explanation and Model Revision

Write your Final Explanation of the driving question as a claim, evidence and reasoning paragraph. Use the numbers from the knives. Then look back at your first model from Lesson 1: say what you had right, what you had to add or change, and which lessons helped you do this.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim and use of the key idea (same force, different area)	States a clear claim that the difference is the area, not the force, uses the words 'same force, different area', and links pressure to force and area correctly.	States that area is different and pressure is greater for the sharp knife, but the idea that force is the same is missing or unclear.	Claim is missing, or says the sharp knife uses more force, or gives no link to pressure.
Calculations and units	All areas and pressures are correct ($5 \times 10^{-6} \text{ m}^2$, $5 \times 10^{-5} \text{ m}^2$, 8 MPa, 0.8 MPa) with correct unit conversions and units throughout the other calculations.	Method is correct but there is one small error in conversion or units.	Several errors or missing units, or $P = F/A$ is not used correctly.
Use of evidence and threshold reasoning	Compares 8 MPa and 0.8 MPa with the 2 MPa value to explain cutting versus squashing, and supports the idea with evidence from the brick, camel and other lessons.	Uses the knife numbers but uses the 2 MPa value or other lessons' evidence only partly.	Gives little or no evidence, or does not use the 2 MPa value.
Extension to liquids, air and hydraulics	Correctly calculates and explains dam pressure, the air force on the glass and the jack, and clearly contrasts the jack (same pressure, different area) with the knives.	Calculates most of these correctly but explanations are brief or the contrast with the knives is weak.	Calculations or explanations are mostly missing or incorrect.
Model revision and communication	Compares the Lesson 1 model with the final model, naming what stayed, what changed and which lessons caused it, in clear organised sentences.	Describes some changes to the model but with few links to lessons.	Does not compare the models or the writing is unclear.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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